DATASCI 185: Computational Methods for Data Science, Fall 2026

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Communications:

Please use the "Discussions" feature to ask questions about course content outside of class time and office hours. This will be monitored by the instructors and the TA's, and you have the option of asking questions anonymously. We prefer this, because multiple students often have the same question, and private emailing limits the number of people helped by an answer. Emails to Dr. Sentz or Dr. Gabriel should be limited to topics pertaining only to you (accomadations, grading questions, etc.)

We will make course announcements through the course Canvas page. **Either turn on Announcement notifications in Canvas, or check it on a regular basis.**

Overview

This course is designed to replace the DATASCI 150-151 computing sequence with a single course. (It will be offered as DATASCI 160 in future semesters). It serves as an introduction to computational methods for students without prior programming experience, to prepare students for upper-level electives in data analysis related courses. While Python is the programming language of choice, the main objective is to understand fundamental programming concepts and to develop computational problem-solving skills. This allows students to develop the foundation necessary to perform data analysis tasks in Python, but also in other programming languages. The course covers functional and object-oriented programming in Python, the development of data science pipelines and workflows, and best practices for debugging code and ensuring robustness. Application datasets are analyzed and visualized using libraries such as Pandas, NumPy, and Matplotlib.

Learning objectives

1. Understand fundamental programming concepts and develop a strong foundation in the Python programming language.
2. Understand Python data types such as integers, floating point numbers, strings, lists, dictionaries, NumPy arrays, and Pandas data frames
3. Apply control flows and functions to solve data science problems computationally.
4. Develop algorithmic problem-solving skills and techniques for debugging code.

5. Learn to utilize software libraries and reproducible workflows to analyze real-world datasets.
6. Communicate computational results effectively using plots and other visualization.

Materials

The class will be entirely based off lectures provided by the instructors for each class and stored in the following Github repository:

https://github.com/sentz2/datasci185fall2026

Optional Materials

The following readings are not required, but are suggested for students interested in further readings:

- Elements of Data Science: https://allendowney.github.io/ElementsOfDataScience/README.html
- Think Python: https://greenteapress.com/thinkpython/html/index.html

Course Schedule

The following table outlines the proposed schedule for the semester. While we intend to follow this, the schedule is **subject to change.** Check Canvas regularly for updates to deadlines and any changes to the assigned material. If changed, homework due dates and quiz dates will **only** be moved to a later time and will never be moved to earlier in the semester.

Date	Title	Lecture	Percentage
	Week 1		
Th 08/27	Syllabus, Intro to Computing Environment	0	
Fr 08/28	Lab 1		
	Week 2		
Tu 09/01	GitHub and Jupyter Notebooks	1	
We 09/02	Assignment 1 due (11:59pm)		5%
Th 09/03	Objects, Variables, and Lists	2	
Fr 09/04	Lab 2		
Fr 09/04	End of Add/Drop/Swap		
	Week 3		
Tu 09/08	Objects, Variables, and Lists (cont.)	3	
Th 09/10	Functions and Libraries	4	
Fr 09/11	Lab 3		
	Week 4		
Mo 09/14	Assignment 2 due (11:59pm)		5%
Tu 09/15	Libraries and Specialized Functions	5	
We 09/16	End of extended drop period		
Th 09/17	Conditionals	6	
Fr 09/18	Quiz 1		6%

	Week 5		
Tu 09/22	Loops	7	
We 09/23	Assignment 3 due (11:59pm)		5%
Th 09/24	Other Compound Data Types	8	
Fr 09/25	Lab 4		
	Week 6		
Tu 09/29	Dictionaries	9	
Th 10/01	User defined functions	10	
Fr 10/02	Quiz 2		6%
	Week 7		
Tu 10/06	More about functions	11	
We 10/07	Assignment 4 due (11:59pm)		5%
Th 10/08	Errors, Exceptions, and Input	12	
Fr 10/09	Lab 5		
	Week 8		
Tu 10/13	No Class (Fall Break)		
Th 10/15	Silent Errors	13	
Fr 10/16	Lab 6		
	Week 9		
Mo 10/19	Assignment 5 due (11:59pm)		5%
Tu 10/20	Classes and Objects	14	
Th 10/22	Classes and Objects (cont.)	15	
Th 10/22	Instructions for Final Project Released		
Fr 10/23	Quiz 3		6%
	Week 10		
Tu 10/27	State Mutation and Inheritance	16	
We 10/28	Assignment 6 due (11:59pm)		5%
We 10/28	Grading basis (Letter vs S/U) and partial course withdrawal (W) deadline		
Th 10/29	APIs, JSON, and Unstructured Data	17	
Th 10/29	Mid-semester survey		0.5%
Fr 10/30	Lab 7		
Fr 10/30	Finalize final project groups		
	Week 11		
Tu 11/03	APIs, JSON, and Unstructured Data (cont.)	18	
We 11/04	Assignment 7 due (11:59pm)		5%
Th 11/05	NumPy and Vectorization	19	
Fr 11/06	Lab 8		

	Week 12		
Tu 11/10	NumPy and Vectorization (cont.)	20	
Th 11/12	Pandas: Subsetting and Filtering	21	
Fr 11/13	Quiz 4		6%
	Week 13		
Tu 11/17	Pandas: Aggregation and Grouping	22	
We 11/18	Assignment 8 due (11:59pm)		5%
Th 11/19	Pandas: Merging and Pivoting	23	
Fr 11/20	Lab 9		
	Week 14		
Tu 11/24	Pandas: Time Series and Panel Data	24	
Th 11/26	No Class (Thanksgiving Break)		
Fr 11/27	No Class (Thanksgiving Break)		
	Week 15		
Tu 12/01	Data Visualization	25	
We 12/02	Assignment 9 due (11:59pm)		5%
Th 12/03	Data Visualization (cont.)	26	
Fr 12/04	Quiz 5		6%
	Week 16		
Mo 12/07	Assignment 10 due (11:59pm)		5%
Tu 12/08	Preview of Algorithms	27	
We 12/09	Final Project due (11:59pm)		20%

Course Assignments

Your grade consists of homework (50%), lab quizzes (30%), and the final group project (20%). The assignments correspond to the structure outlined in the schedule above.

Course Assignments Description

Homework Assignments

Working together on the homework assignments is encouraged, but you must write your own solutions. It is highly recommended that you make your solo effort on all the problems before consulting others. Each assignment has its due date indicated in this Syllabus and on Canvas, and will be due at 11:59pm on that date.

Late homework submissions will receive a 5% penalty for every 12 hour period after the due date. After 5 days (120 hours), submissions will receive no credit. Percentages are calculated with respect to the total number of points on the homework (5 points). For clarity, here is the explicit policy:

- 1 minute – 12 hours late: 5% deduction
- 12 hours – 24 hours late: 10% deduction
- 24 hours – 36 hours late: 15% deduction

- 36 hours – 48 hours late: 20% deduction
- 48 hours – 60 hours late: 25% deduction
- 60 hours – 72 hours late: 30% deduction
- 72 hours – 84 hours late: 35% deduction
- 84 hours – 96 hours late: 40% deduction
- 96 hours – 108 hours late: 45% deduction
- 108 hours – 120 hours late: 50% deduction
- More than 120 hours late: no credit

To accommodate unexpected circumstances, **your lowest homework grade will be automatically dropped at the end of the semester.**

Unpenalized homework extensions may be given in the following circumstances:

- Students who are approved for flexibility with assignment due dates through the Office of Accessibility services may be given extensions in cases dictated by *The Flexibility with Assignment Due Dates Agreement* form.
- In the following cases, **where students have contacted the Office for Undergraduate Education in accordance with Emory policies:**
    - Bereavement
    - Illness causing a four calendar-day absence (or longer) that is supported by medical documentation
    - Situations where students are working with campus officials to resolve ongoing circumstances beyond their control
    - Student is representing the University at an officially sanctioned event.

Homework will be submitted on Canvas through the Gradescope tool. In case students have connection issues with Gradescope, they should be prepared to contact one of us through email **before the deadline** and **with their homework submission attached.**

Submission policies: It is your responsibility to submit homework in the correct format and to ensure you have submitted the correct file.

- If the homework file has the wrong format, but your solution is visible after downloading the file, your score will be reduced by 10%
- If you submit an unrelated file, or a blank or corrupted file, you will be given another chance to submit with a 25% reduction. You will only be given this chance the first time this mistake is made. Any additional instances will receive a zero.

In-class quizzes

Five in-class quizzes will be given throughout the semester. They will be given on the dates listed in the schedule above, except for times when the instructor announces a change in advance. **Quizzes will never be moved to an earlier date.**

The quizzes will last the entire 50 minute lab period, and students should not collaborate with others. The quizzes will be open note, but students cannot use internet resources outside the course Canvas page and the course GitHub repo. **No AI/LLM use is allowed during quizzes**.

Students *must* submit their quiz to Canvas while they are present in the classroom. Submitting a quiz after leaving the classroom is a violation of Emory's Undergraduate Academic Honor Code and will be reported to the Honor Council.

Rescheduling Quizzes: To discourage students coming to class sick, I will allow students to take the quiz at another time if they notify us before Noon on the date of the quiz. This is up to our discretion, however. Keep in mind that intentionally giving false information to an instructor is a violation of Emory's Academic Honor Code and will be reported to the Honor Council.

Your lowest quiz grade will be automatically dropped at the end of the semester.

Submission policies: The submission policies for quizzes are identical to the submission policies for homework given above.

Final Project

The final term project is group-based, in teams of 3-4 students. The details of the project will be discussed later in the semester.

Participation (extra credit) + 1%

- Your participation will consist of a midsemester in-class survey. This will be worth 0.5% extra credit.
- You will also be granted 0.5% extra credit if you fill out the final course evaluations.

Grading Scale

Your final grade will be assigned based on the grading scale below:

A	A-	B+	B	B-	C+	C	C-	D	F
93+	90-92	87-89	83-86	80-82	77-79	73-76	70-72	60-69	0-59

Grades will be rounded to the nearest integer (a 92.5 will be rounded to a 93, while a 92.4 will be rounded to a 92). Throughout the semester, I will keep the Canvas gradebook up to date so that you have an ongoing understanding of where your grades fall in this scale.

Policies

General: Students are expected to adhere to the Emory College Honor Code as well as its Conduct Code, see https://catalog.college.emory.edu/policies/honor-code.html. Specifically, the honor code is in effect throughout the semester. By taking this course, you affirm that it is a violation of the code to cheat on assignments, end-of-chapter assessments, to plagiarize, to deviate from the teacher's instructions about collaboration on work that is submitted for grades, to give false information to a faculty member, and to undertake any other form of academic misconduct. You also affirm that if you witness others violating the code you have a duty to report them to the honor council.

<u>AI Policy:</u> Students should NOT use ChatGPT/Co-pilot or any other LLM tools when working on assignments. Working through problems and correcting errors yourself will pay off in the long run.

According to Emory Undergraduate Academic Honor Code, the use of AI to generate any content for an assignment constitutes plagiarism unless the student appropriately acknowledges in the assignment the extent to which an artificial intelligence program contributed to the work.

It is considered an academic integrity violation to submit any part of a homework problem as a prompt to LLMs.

AI/LLM use on quizzes is **strictly prohibited.** Students who are found to have used AI/LLM assistance on a quiz will be reported to the Emory College Honor Council.

Using AI tools in a manner prohibited in this course syllabus constitutes Cheating under the Emory Honor Code and is thus a form of academic misconduct.

<u>Special circumstances:</u> Students requiring any type of special classroom/testing accommodation for a disability, religious belief, scheduling conflict, or other impairment that might affect his or her successful completion of this course must personally present the requested remedy or other adjustment in written form (signed and dated) to the instructor, i.e., supporting memorandum of accommodation from the Department of Accessibility Services, https://accessibility.emory.edu/index.html. Requests for accommodations must be received and authorized by the instructor in written form no less than two weeks in advance of need. No accommodation should be assumed unless so authorized. In the event of needs identified later in the course, or for which an adjustment cannot be made on a timely basis, a grade of "I" Incomplete for the course, will be given to accommodate the unanticipated request.

<u>Attendance:</u>

- There is no separate grade for attendance. However, attendance will be monitored through quizzes which will take place during class.
- The OUE student self-service absence form must be submitted if you would like to have your absence excused due to the reasons recognized by the university (See the top of the form). Be sure to complete this form in advance, and you must include all classes you are currently enrolled at the time of completing the form so that all your instructors will be notified all at once. If you wish to explain your situation further, you may also email your instructor in advance, and if necessary, submit official documentations.

<u>In/Out of classroom conduct:</u> Students are expected to adhere to the Emory University Code of Conduct, see https://conduct.emory.edu/policies/codes.html